Is Financial Volatility Intrinsically Rough? A Critical Review of Estimation Methods and Empirical Evidence*

Konstantin Iliev, Dimitar Mitov
Delft University of Technology, Faculty of EEMCS

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Abstract

The empirical discovery that log-volatility behaves like fractional Brownian motion with a Hurst exponent near 0.1 has motivated rough fractional stochastic volatility models. However, whether this roughness is an intrinsic property of instantaneous volatility or a byproduct of sampling frequency, volatility proxy construction, and estimator choice remains disputed. This paper reviews the debate through a comparative literature review of four primary studies: Gatheral et al. [2], Fukasawa et al. [4], Bolko et al. [5], and Cont and Das [3]. The papers are compared across four main points: the volatility measure used, the roughness estimation procedure, the treatment of estimation error, and the resulting empirical conclusions. Gatheral et al. [2] infer roughness from scaling behaviour of volatility proxies; Fukasawa et al. [4] and Bolko et al. [5] develop estimators that correct for realized-volatility measurement error and continue to support rough volatility under their respective models; Cont and Das [3] show nonparametric counterexamples in which non-rough processes generate apparently rough realized-volatility signals. The review finds that methodological differences - particularly sampling frequency, proxy construction, and error-correction assumptions - explain part of the apparent disagreement. However, corrected estimators rely on specific model assumptions, and the intrinsic roughness of instantaneous volatility is not fully established. Further research comparing estimators under common simulation conditions is needed.

1 Introduction

The mathematical modelling of stock market derivatives is commonly described using continuous-time stochastic differential equations:

$$dV = \mu\, dt + \sigma\, dW$$

where μ denotes the drift, σ denotes the volatility, and W is a Wiener process. Volatility has a substantial effect on derivative prices, making it an important parameter to trace, estimate, model, and, above all, forecast.

There have been numerous attempts at doing so. For example, an assumption of the Black-Scholes model is that the volatility term is constant [1]. Many other models have considered alternative representations of volatility. More recently, local-stochastic-volatility models have been developed to fit observed market prices while generating plausible volatility dynamics [2].

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The different possible representations of volatility over time imply different levels of smoothness of the volatility curve. While direct explicit functional substitution offers a clear but potentially market-inconsistent result, models incorporating long-memory effects have also been introduced to account for persistent volatility dynamics. The Fractional Stochastic Volatility model (FSV) proposes a sequence of autocorrelated values in a Gaussian process, described by:

$$\mathbb{E}\left[\left|W_{t+\Delta}^{H} - W_{t}^{H}\right|^{q}\right] = K_q \Delta^{qH}$$

whereby W^H is a fractional Brownian motion, t is non-dimensional time, Δ the time step, q the order of the absolute moment, K_q the q-th absolute moment of a standard normal variable, and H is the Hurst coefficient, which is a measure of the roughness of the process. Classical FSV specifications typically consider values of $H > 0.5$, corresponding to persistent and comparatively smooth volatility dynamics. However, Gatheral et al. [2] revisit the smoothness of volatility using high-frequency data and find that log-volatility behaves approximately like fractional Brownian motion with $H \approx 0.1$. This motivates their Rough Fractional Stochastic Volatility model:

$$\mathbb{E}\left[|\log(\sigma_\Delta) - \log(\sigma_0)|^{q}\right] = K_q \nu^{q} \Delta^{qH}$$

that uses the same notation and introduces ν as the so-called intermittency parameter aimed at measuring the intensity of the observed stochastic processes.

Studies have already been performed on the topic, and there is no full scientific consensus on whether the observed roughness of volatility is an intrinsic property of the volatility process, with Cont and Das [3] arguing against volatility roughness, while Gatheral et al. [2], Fukasawa et al. [4], and Bolko et al. [5] provide evidence supporting rough-volatility specifications. The aim of this paper is to review whether volatility roughness is intrinsic or in part or fully a byproduct of sampling and selection of analytic methodology.

The rest of the paper is structured as follows: section 2 presents a comparative literature review method, including the criteria used to select the main studies and parameters along which they are compared. Onwards, section 3 provides the theoretical background of the study. Section 4 presents the results, first reviewing the initial evidence for rough volatility, then examining estimators that correct for measurement error, and finally considering counterexamples that offer an alternative viewpoint. Section 5 discusses the methodological sources of the apparent disagreement, assesses the reliability and limitations of the reviewed approaches, and identifies directions for future research. Section 6 concludes the literature review.

2 Method

This paper is a comparative literature review investigating the effects of differences in methodology on the empirical evidence for roughness of volatility. ChatGPT was asked to suggest starting papers through the prompt "suggest me papers discussing volatility roughness", though those were only opened subsequently throught other websites as ChatGPT sources are often dead-end connections. Relevant literature was gathered through targeted searches (including typing in the titles suggested by ChatGPT) of publisher databases and academic repositories, including arXiv, SSRN, SpringerLink, Wiley Online Library, and ScienceDirect. Search terms included “fractional stochastic volatility”, “rough fractional stochastic volatility”, “rough volatility”, “Hurst exponent volatility”, “realized volatility roughness”, and “volatility roughness estimation error”.

Studies were considered relevant if they investigated the empirical roughness of financial volatility, proposed methods for estimating the Hurst parameter, or examined the influence of volatility estimation errors on roughness measurements. Studies without sufficient methodological or empirical information for critical comparison were excluded. Four primary studies were selected: Gatheral et al. [2], Fukasawa et al. [4], Bolko et al. [5], and Cont and Das [3]. These studies were chosen because they address different aspects of the same methodological debate and

explicitly engage with findings or limitations identified in earlier research. Additional academic sources provide theoretical background and contextual information.

The selected studies are compared across four criteria. First, the volatility measure used is identified, distinguishing between instantaneous volatility, integrated variance, and realized-volatility proxies. Second, the roughness estimation procedure is examined, including its mathematical formulation and fundamental assumptions. Third, the treatment of estimation error is evaluated by identifying the sources of error considered and the corrections or robustness analyses employed. Finally, the empirical findings are compared in relation to the conclusions they support about the underlying volatility process.

Particular attention is given to whether differences in estimation methods explain seemingly conflicting results. Findings are compared within the context of their statistical assumptions, rather than treating estimated Hurst parameters as directly interchangeable.

This review is limited by its targeted, non-exhaustive literature selection and its reliance on results reported in the original studies. Furthermore, differences in datasets, model specifications, and observation procedures restrict direct quantitative comparison. The review therefore focuses on evaluating methodological assumptions, interpreting reported evidence, and identifying remaining limitations rather than independently validating the estimators.

3 Theoretical Background

3.1 Fractional Stochastic Volatility and the Hurst Exponent

In stochastic volatility models, the volatility of an asset is itself represented by a stochastic process. Fractional stochastic volatility (FSV) models extend this framework by incorporating fractional Brownian motion, denoted by B_t^H, where $H \in (0,1)$ is the Hurst exponent. The increments of fractional Brownian motion satisfy

$$\mathbb{E}\left[\left|B_{t+\Delta}^H - B_t^H\right|^q\right] = K_q \Delta^{qH}, \tag{1}$$

where Δ denotes the time lag, $q > 0$ is the moment order, and K_q is the corresponding absolute moment of a standard normal random variable.

The Hurst exponent characterizes the local regularity of fractional Brownian motion. For $H = 0.5$, the process reduces to standard Brownian motion. Values $H < 0.5$ correspond to rougher sample paths, while $H > 0.5$ corresponds to comparatively smoother paths. Within the fractional Brownian framework, the Hurst exponent also determines the dependence structure of increments, although these properties need not be linked by a single parameter in more general stochastic processes.

Classical FSV specifications typically consider $H > 0.5$, while rough fractional stochastic volatility (RFSV) models use $H < 0.5$. Gatheral et al. [2] motivate the latter by finding empirical scaling behaviour consistent with $H \approx 0.1$. The distinction is therefore central to determining whether observed volatility dynamics are intrinsically rough or merely appear rough due to the estimation procedure.

3.2 Instantaneous, Integrated, and Realized Volatility

Consider an asset price S_t satisfying

$$\frac{dS_t}{S_t} = \mu_t \, dt + \sigma_t \, dW_t, \tag{2}$$

where μ_t denotes the drift, σ_t the instantaneous volatility, and W_t a standard Brownian motion. Although asset prices can be observed, the instantaneous volatility process σ_t is latent and must be estimated from financial data.

An important distinction exists between instantaneous variance, σ_t^2, and integrated variance, defined over an interval $[t, t+\Delta]$ as

$$IV_{t,t+\Delta} = \int_t^{t+\Delta} \sigma_s^2 \, ds. \tag{3}$$

Integrated variance describes the accumulated variance over an observation interval rather than its value at a particular instant. In practice, it is commonly estimated using realized variance, constructed from discrete logarithmic price returns:

$$RV_{t,t+\Delta}^{(n)} = \sum_{j=1}^{n} \left(\log S_{t_j} - \log S_{t_{j-1}}\right)^2, \tag{4}$$

where $t = t_0 < t_1 < \cdots < t_n = t + \Delta$. Under suitable assumptions, including continuous asset prices and the absence of market microstructure noise, realized variance converges in probability to integrated variance as the observation frequency increases.

However, realized variance is not a direct observation of instantaneous variance. Finite sampling introduces estimation error, while integration over time alters the observed fluctuations of the volatility process. Additional distortions may arise from market microstructure noise and the choice of sampling frequency.

Consequently, the regularity of a realized-volatility series doesn't have to coincide with that of the underlying instantaneous volatility. This distinction is central to the methodological disagreement examined in Sections 4 and 5, because different estimators may capture different properties of the latent process.

4 Results

4.1 Initial Empirical Evidence for Rough Volatility

Gatheral et al. [2] provide the initial empirical evidence motivating rough fractional stochastic volatility (RFSV) models. Challenging the conventional fractional stochastic volatility assumption of $H > 0.5$, they investigate the scaling behaviour of log-volatility increments across different time intervals. Their analysis considers DAX and Bund futures, for which volatility proxies are constructed from high-frequency estimates of integrated variance, and the S&P and NASDAQ indices, for which daily realized-variance estimates are used.

To quantify volatility roughness, the authors calculate the empirical q-th absolute moment of log-volatility increments:

$$m(q, \Delta) = \frac{1}{N} \sum_{k=1}^{N} \left|\log(\sigma_{k\Delta}) - \log(\sigma_{(k-1)\Delta})\right|^q, \tag{5}$$

where σ denotes the volatility proxy, Δ is the observation lag, and q is the moment order. Under fractional scaling, these moments satisfy approximately

$$m(q, \Delta) \propto \Delta^{qH}. \tag{6}$$

Consequently, the Hurst parameter can be estimated through a linear regression of $\log m(q, \Delta)$ against $\log \Delta$, the slope of which is expected to equal qH. By repeating this procedure for different values of q, the authors analyse whether the estimated scaling exponents are consistent with a common Hurst parameter.

The empirical results exhibit approximately linear scaling relationships, yielding Hurst estimates of 0.125 for DAX, 0.082 for Bund, 0.142 for S&P, and 0.139 for NASDAQ. These values are significantly below the Brownian-motion benchmark of $H = 0.5$. These results suggest that

observed volatility is considerably rougher than assumed in classical fractional stochastic volatility models. Paired with the approximately Gaussian distribution of log-volatility increments, these findings motivate the RFSV specification with $H < 0.5$.

However, these estimates are obtained from volatility proxies and are not directly observed instantaneous volatility. Therefore one cannot consider this proof of intrinsic rough volatility without considering how the volatility estimation procedure, including sampling error and temporal aggregation, distort the scaling behaviour of the underlying process. This distinction becomes central to subsequent studies examining the effects of volatility estimation errors.

4.2 Measurement Error and Corrected Roughness Estimation

The main challenge in interpreting the results of Gatheral et al. [2] is that instantaneous volatility cannot be observed directly. Fukasawa et al. [4] investigate whether estimation errors in realized volatility can themselves generate the scaling behaviour associated with a small Hurst parameter. In simulations described in an earlier version of their study [6], they generate asset prices with Brownian-driven log-volatility, for which the true value is $H = 0.5$. Applying the original log-log regression to realized volatility yields an apparent $H = 0.4016$ with one-second return sampling, but only $H = 0.0930$ with five-minute sampling. Thus, the same latent process can appear substantially rougher simply because the volatility proxy is constructed at a different sampling frequency. These simulations indicated that a convincing linear scaling relationship is not necessarily a reliable estimate of latent volatility roughness.

To address this problem, Fukasawa et al. [4] develop a quasi-likelihood estimator for an approximately log-normal fractional stochastic volatility model. Rather than estimating H solely from the slope of empirical increment moments, they model the distribution of the realized-volatility estimation error and combine it with a Whittle-type approximation to the covariance of latent log-volatility. Their approach uses the dependence structure of the observations to separate variation attributed to the fractional process from variation introduced by volatility estimation. They establish consistency under high-frequency asymptotics and assess finite-sample behaviour through simulations. Their empirical analysis continues to indicate $H < 0.5$, despite explicitly accounting for sampling error. However, this conclusion is conditional on the assumed fractional model and the observation-error approximation; consistency within this framework does not establish that the true market process belongs to the same model class.

Bolko et al. [5] address sampling error through a different statistical construction. Their generalized method of moments (GMM) estimator matches theoretical moments and autocovariances of *integrated variance* to their empirical counterparts. In practice, integrated variance is replaced by realized variance computed from intraday returns. This substitution introduces sampling error: in particular, the observed second moment contains an additional error-variance contribution. If the moment conditions derived for integrated variance are applied unchanged to realized variance, this contribution may be misinterpreted as a property of the latent volatility dynamics, biasing estimates of H downwards. The authors therefore introduce an analytical correction into the GMM moment conditions, including an implementation based on integrated-quarticity information.

The correction has a measurable effect in their simulations. When the true Hurst parameter is $H = 0.5$, one reported configuration gives an average estimate of 0.2725 using uncorrected realized variance, compared with 0.4912 after correction [5]. The corrected estimator therefore comes substantially closer to the true value in this setting. Applied to realized-variance series from major equity indices, their method nevertheless yields estimates around $H = 0.05$, supporting rough volatility within the fitted fractional log-normal framework.

Although both studies correct for volatility-proxy error, their methods are not equivalent. Fukasawa et al. use a quasi-likelihood based on short-interval asymptotics for log-realized volatility, whereas Bolko et al. fit a parameterized model to moments of integrated variance and explicitly correct finite-sample moment distortion. Bolko et al. also question the correspondence

between the short-interval approximation in Fukasawa's theory and its application to daily realized measures. Their similar empirical conclusions therefore do not amount to two estimates obtained under identical assumptions. Both studies show why uncorrected roughness measures can be misleading, while leaving open how robust their corrected inferences are to alternative latent models and more general observation errors.

4.3 Counterexamples and Challenges to Intrinsic Roughness

Cont and Das [3] challenge the interpretation of empirical roughness as an intrinsic property of instantaneous volatility. Rather than assuming a fractional stochastic volatility model, they introduce a nonparametric roughness estimator based on normalized p-th variation. For fractional Brownian motion, the critical variation exponent satisfies $p^* = 1/H$, allowing the Hurst parameter to be inferred from the regularity of an observed path. Their approach therefore assesses roughness without requiring a parametric specification of the volatility dynamics.

To examine whether realized volatility accurately reflects the roughness of the underlying process, the authors simulate asset prices using stochastic volatility models with known Hurst parameters. One example uses a mean-reverting Ornstein-Uhlenbeck (OU) log-volatility process:

$$\begin{aligned} dS_t &= S_t \sigma_t \, dB_t, \\ \sigma_t &= \sigma_0 e^{Y_t}, \\ dY_t &= -\gamma Y_t \, dt + \theta \, dW_t, \end{aligned} \tag{7}$$

where B_t and W_t are independent Brownian motions. Consequently, the instantaneous volatility has Brownian path regularity, corresponding to $H = 0.5$. The authors compare roughness estimates obtained directly from the simulated instantaneous volatility with those calculated from realized volatility constructed using discrete price observations.

Across 2500 simulated paths, the average estimated roughness index is approximately 0.55 for instantaneous volatility, but only 0.13 for realized volatility [3]. The latter closely resembles empirical estimates commonly interpreted as evidence for rough volatility, despite the process having $H = 0.5$. The authors extend this analysis to fractional OU models with different Hurst parameters. For example, when the true parameter is $H = 0.8$, the estimated roughness index of realized volatility is only 0.052. These results demonstrate that realized-volatility roughness need not reflect the regularity of the process.

Cont and Das further examine the errors introduced by volatility estimation. Their simulations reveal that these errors can exhibit complex temporal dependence, which may not be adequately represented by independent measurement noise. However, they also observe that log-volatility estimation errors in their OU example are closer to independent, consistent with the approximation of Fukasawa et al.'s approach. The validity of such error assumptions therefore depends on the volatility model and observation scheme.

These counterexamples establish that a low roughness estimate obtained from realized volatility does not necessarily imply intrinsically rough instantaneous volatility. Nevertheless, they do not demonstrate that actual market volatility is smooth, nor do they directly invalidate the corrected estimators of Fukasawa et al. [4] and Bolko et al. [5]. Instead, they highlight the importance of distinguishing the roughness of an observed volatility proxy from inference about the latent process, particularly when comparing results obtained under different methodological assumptions.

4.4 Comparative Summary of the Evidence

The four studies differ in their volatility measures, estimation procedures, and treatment of measurement error. Table 1 summarizes these methodological differences and their implications for interpreting empirical volatility roughness.

Table 1: Comparison of approaches to volatility roughness estimation

Study	Estimation approach	Treatment of error	Main finding
Gatheral et al. (2018)	Log-log scaling of estimated log-volatility increments	Examines aggregation effects, but does not provide a general measurement-error correction	Volatility proxies exhibit rough scaling with $H \approx 0.1$
Fukasawa et al. (2022)	Quasi-likelihood estimation using realized volatility	Models realized-volatility estimation error	Corrected estimates continue to support $H < 0.5$
Bolko et al. (2023)	GMM using moments of integrated variance	Introduces an analytical correction for realized-variance sampling error	Corrected estimates suggest $H \approx 0.05$
Cont and Das (2024)	Nonparametric roughness estimation using normalized p-variation	Examines estimation-induced roughness through simulations	Non-rough spot volatility can produce apparently rough realized volatility

A central distinction concerns the object of inference. Gatheral et al. [2] identify rough scaling in empirical volatility proxies, whereas Fukasawa et al. [4] and Bolko et al. [5] estimate the Hurst parameter within specified fractional volatility models. Both corrected approaches continue to support rough volatility, suggesting that sampling error alone does not explain their empirical findings under the assumed models.

Cont and Das [3], however, demonstrate that the roughness of a realized-volatility signal need not coincide with that of instantaneous volatility. Their counterexamples challenge the direct interpretation of proxy roughness, but do not establish that the corrected estimators are invalid.

Consequently, the findings are not necessarily contradictory. While estimation errors can generate artificial roughness, model-based corrections may still provide evidence for genuinely rough volatility. The remaining uncertainty concerns whether these conclusions remain reliable when the assumptions used for the estimation methods are violated.

5 Discussion

5.1 Sources of the Apparent Disagreement

The apparent disagreement regarding intrinsic volatility roughness can be attributed to three main methodological differences: the volatility measure used, the treatment of estimation error, and the assumptions underlying roughness estimation.

First, the studies do not necessarily examine the same statistical object. Gatheral et al. [2] infer roughness from the scaling behaviour of empirical volatility proxies, while Fukasawa et al. [4] and Bolko et al. [5] estimate parameters of latent fractional volatility models using realized or integrated variance. Cont and Das [3], in contrast, investigate whether the pathwise roughness of realized volatility reflects that of instantaneous volatility. Consequently, their estimates are not necessarily directly comparable.

Second, the studies differ in their treatment of measurement error. Fukasawa et al. and Bolko et al. explicitly incorporate estimation errors into their respective estimation procedures, whereas Cont and Das demonstrate that such errors can exhibit dependence structures that complicate statistical inference. The effectiveness of an error correction therefore depends on whether its assumptions adequately describe the observation process.

Finally, the studies differ in their modelling assumptions. Fukasawa et al. and Bolko et al. estimate roughness within specified fractional volatility models, while Cont and Das employ a nonparametric approach. Although the corrected parametric estimates support rough volatility, their interpretation depends on the validity of the underlying model specifications.

These distinctions suggest that the conflicting findings are not necessarily mutually exclusive. The corrected estimators provide evidence for rough volatility under particular assumptions,

while the counterexamples demonstrate that roughness observed in a volatility proxy does not independently establish intrinsic roughness. The disagreement therefore concerns not only the estimated Hurst exponent, but also the conditions under which it can be interpreted as a property of the latent volatility process.

5.2 Reliability and Limitations

The reliability of empirical roughness estimates depends on whether the estimation procedure can distinguish fluctuations of the underlying volatility process from those introduced by its observation. Consequently, the evidence provided by each study must be interpreted in relation to the assumptions and limitations of its methodology.

The scaling relationships identified by Gatheral et al. [2] provide evidence that empirical volatility proxies exhibit rough behaviour across several financial assets. However, these findings alone cannot establish whether the estimated roughness originates from the instantaneous volatility process or from its estimation.

Fukasawa et al. [4] and Bolko et al. [5] address this limitation by developing estimators that explicitly account for realized-volatility measurement error. Their theoretical results and simulation studies demonstrate that these methods can reduce estimation bias under specified conditions. Moreover, both studies continue to obtain low Hurst estimates after applying their corrections, providing additional evidence for rough volatility. Nevertheless, the reliability of these conclusions depends on the validity of their fractional volatility models and assumptions regarding the observation errors.

Cont and Das [3] highlight the importance of these limitations by demonstrating that non-rough volatility processes can generate apparently rough realized-volatility signals. Their counterexamples establish that low empirical roughness estimates are not sufficient to identify intrinsic roughness. At the same time, these results do not directly contradict the consistency properties established for the corrected estimators under their respective assumptions.

Overall, the literature provides evidence that rough volatility models remain compatible with empirical observations after accounting for certain estimation errors. However, this does not establish that instantaneous volatility is intrinsically rough independently of the assumed model and observation procedure. The remaining uncertainty concerns the robustness of these conclusions when the underlying statistical assumptions do not hold.

Although existing studies have demonstrated the influence of volatility estimation errors and developed methods to correct for them, uncertainty remains regarding the robustness of roughness estimates under different statistical assumptions. In particular, the corrected estimators of Fukasawa et al. [4] and Bolko et al. [5] rely on specific volatility models and observation-error assumptions. The extent to which their conclusions remain valid when these assumptions are violated requires further investigation.

One possible direction is a systematic comparison of roughness estimators under identical simulation conditions. By generating financial price data from volatility processes with known Hurst parameters, different estimation methods could be evaluated using the same observations. Varying the sampling frequency, volatility estimation window, and dependence structure of measurement errors would help identify the conditions under which each method produces reliable estimates.

Furthermore, testing these methods on volatility processes outside their assumed model families could reveal their sensitivity to model misspecification. Such comparisons should examine not only estimation bias, but also whether the methods can distinguish genuinely rough volatility from non-rough processes exhibiting apparent roughness.

Finally, applying alternative estimation methods and nonparametric tests to the same empirical datasets could help establish which findings are robust to methodological choices. Although the true instantaneous volatility cannot be directly observed in financial markets, agreement across methods with different assumptions would strengthen the empirical evidence.

These investigations could help clarify whether the remaining disagreement reflects limitations of existing estimation procedures or genuine uncertainty about the regularity of financial volatility.

6 Conclusion

This review investigated to what extent methodological differences in volatility estimation and roughness measurement explain the conflicting evidence on the intrinsic roughness of financial volatility.

The reviewed studies employ different volatility proxies, estimation procedures, and statistical assumptions. Gatheral et al. [2] infer roughness from the scaling behaviour of estimated volatility, while Fukasawa et al. [4] and Bolko et al. [5] estimate fractional volatility parameters using methods that account for measurement error. Cont and Das [3] instead examine the relationship between the pathwise roughness of realized and instantaneous volatility.

The findings demonstrate that volatility estimation errors can generate artificially low Hurst estimates, even when the underlying process is not rough. Although the corrections developed by Fukasawa et al. and Bolko et al. address particular sources of estimation bias and continue to support rough volatility, their conclusions depend on the validity of their modelling assumptions. The counterexamples of Cont and Das demonstrate that apparent roughness in volatility proxies does not necessarily imply intrinsic roughness.

Methodological differences therefore explain an important part of the apparent disagreement, as the studies do not always estimate the same quantity or establish conclusions under comparable conditions. Simultaneously, these differences do not fully resolve the question of intrinsic volatility roughness. Further research comparing estimation methods under common simulation conditions, varying measurement errors and model assumptions, is needed to establish how reliably empirical observations identify the regularity of the latent volatility process.

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A Use of Generative AI

During the preparation of this work, the authors used DeepSeek to format text and grammar and align citations with the appropriate format. ChatGPT was asked to suggest titles, but those were extracted from academic websites, due to ChatGPT knowing the titles but being unable to provide the PDF files themselves. After using these tools, the authors reviewed and edited the content to assure the meaning remains unchanged responsibility for the content of the article.

Table 2 summarizes the generative AI usage described above. This table was generated by DeepSeek using the prompt ".

Table 2: Generative AI tools used, their purposes, and prompts.

Tool	**Section(s)**	**Purpose**	**Prompt(s)**
ChatGPT	Method (Literature search)	Suggested starting paper titles on volatility roughness. These titles were subsequently located and verified through academic websites and publisher databases.	"suggest me papers discussing volatility roughness"
DeepSeek	Abstract, Sections 1–6, References, Appendix	Formatted text and LaTeX, checked grammar, aligned citations with IEEE numeric style, and generated the AI-use summary table.	"Format the latex file" "Align the citations with IEEE" "Generate a table summarizing AI use [screenshot of the task]" "Fix the abstract style to IEEE"

AI is helpful in repetitive, routine tasks, and does them reliably. For the initial search, it does a good job identifying titles quickly and is helpful as a starting point. Simultaneously, its text-writing is typically overcomplicated and can often be seen restating claims without adding new information. For that reason, the team's AI use was limited to formatting and initial-search prompting.

Initially, the team agreed on minimal AI use as it spawns complex sentences that are too hard to stomach on one's own without further explanation and steer one too much into a certain direction, not allowing for an original authorial outlook. Those were followed through on along the project.

For the intended use of paper generation and citations, DeepSeek and ChatGPT fulfilled their tasks fully as envisioned.